

MELODY XUAN LIM, PH.D

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EDUCATION

University of Chicago

PhD in Physics

Advisor: Heinrich M. Jaeger

Thesis: Acoustically Levitated Granular Matter

Chicago, IL

August 2016 – March 2022

Duke University

BSc in Physics

Summa Cum Laude

Graduation with Distinction

Durham, NC

August 2012 – May 2016

ACADEMIC EMPLOYMENT

Cornell University

Postdoctoral Research Associate

Department of Physics

Laboratory of Atomic and Solid State Physics

Supervisors: Itai Cohen, Paul McEuen

Funded by: KIC Experimental Postdoctoral Fellowship

Ithaca, NY

March 2022 – present

AWARDS AND HONORS

Academic Awards

- Whetten Memorial Award, Cornell Nanoscale Facility 2023
In recognition of outstanding young women in science and engineering whose research was conducted in the CNF.
- KIC Experimental Postdoctoral Fellowship, Cornell University 2022
In support of the best and brightest young researchers in nanoscale science to pursue postdoctoral work at Cornell.
- Yodh Prize, University of Chicago 2021
In recognition of outstanding experimental work by a graduate student in physics
- Soft Matter for All Symposium 2020
In promotion of early-career researchers in the field of Soft and Living Matter
- Eckhardt Scholar, University of Chicago 2016–2022
Awarded to support exceptional doctoral students
- Maria Goeppert-Mayer fellowship, University of Chicago 2016–2017
- Sachs fellowship, University of Chicago 2016–2017
- Phi Beta Kappa member
- Daphne Chang Memorial Award, Duke University 2015
Awarded to students who excel in undergraduate research
- Lemelson-MIT Student Prize Finalist 2016

Travel & Speaker Awards

- Future Faculty Symposium Travel Award, Society of Engineering Science 2025
- Best poster, New York Nanoscale Symposium 2023
- Best talk, Research Triangle CUWiP 2015
- Poster Award, 1st prize, Duke University undergraduate presentation 2010, 2011

INVITED TALKS

American Physical Society March Meeting	2026
Society of Engineering Science Annual Technical Meeting	2025
Hong Kong University of Science and Technology, Physics Department	2025
Cornell University Soft Matter Seminar	2025
Syracuse University Soft Matter and Biophysics Seminar	2025
American Physical Society March Meeting	2022
Cornell University Kavli Institute Special Seminar	2022
Soft Matter for All Symposium	2020
James Franck Institute Friday Seminar	2020
James Franck Institute Friday Seminar	2019
Granular Matter Gordon Research Conference	2018
James Franck Institute Friday Seminar	2018

TEACHING, OUTREACH, AND SERVICE

Mentees

· Winter Knutson, undergraduate student	2024
· Naomi Naranjo, undergraduate student	2023
· Brady Wu, first year graduate student	2020–2021
· Annie Xia, undergraduate student	2019–2026
· Tali Khain, first year graduate student	2019
· Tchakamau Ra, postbac researcher	2019–2021
· Adam Kline, first year graduate student	2018–2019
· Isaac Harris, undergraduate student	2017–2019

Journal Review

· Europhysics Letters, Granular Matter, Applied Physics Letters, Astronomy and Astrophysics, Experimental Astronomy, Sensors and Actuators, Journal of the Acoustical Society of America	
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Outreach

· Cornell Nano Day	2024
· Cornell Expanding Your Horizons, Workshop Leader	2024
· University of Chicago annual open house, “Physics with a Bang”	2016–2019

Service to Professional Societies

· Session organizer, Division of Nonlinear and Statistical Physics, American Physical Society	2026
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LIST OF PUBLICATIONS

Peer-reviewed journal articles

- Liang, Z.* , **Lim, M. X.***, Zhu, Q, Mottes, F., Kim, J. Z., Guttieres, L., Smart, C., Pearson, T., Du, C. X., Brenner, M. P., McEuen, P. L., & Cohen, I. Magnetic Decoupling for High-Yield, Time-Efficient Microscale Self-Assembly, Proceedings of the National Academy of Science, 122 (35) e2502361122. (* indicates equal contribution).
- C.L. Smart, T. G. Pearson, Z.Liang, **M.X. Lim**, M.I. Abdelrahman, F. Monticone, I. Cohen, P.L. McEuen. Magnetically programmed diffractive robotics , Science **386**(6725), 2024.
- B. Wu, B. vanSaders, **M.X. Lim**, H.M. Jaeger. Hydrodynamic coupling melts acoustically levitated crystalline rafts, Proceedings of the National Academy of Science, 120 (29) e2301625120.
- **M.X. Lim**, H.M. Jaeger. Acoustically levitated lock and key grains, Phys. Rev. Research, **5**, 013116.
- **M.X. Lim**, B. vanSaders, A. Souslov, H.M. Jaeger. Mechanical properties of acoustically levitated granular rafts, Phys. Rev. X, **12**, 021017.
- **M.X. Lim**, K.A. Murphy, H.M. Jaeger. Edges control clustering in levitated granular matter. Granular Matter, **21**(3), 77, 2019
- **M.X. Lim**, A. Souslov, V. Vitelli, H.M. Jaeger. Cluster formation by acoustic forces and active fluctuations in levitated granular matter. Nature Physics, **15**(5), 460–464, 2019
- **M.X. Lim**, R.P. Behringer. Topology of force networks in granular media under impact. EPL, **120**(4), 44003, 2018
- **M.X. Lim**, J. Barés, H. Zheng, R.P. Behringer. Force and mass dynamics in non-Newtonian suspensions. Physical Review Letters **119**(18), 184501, 2017
- **M.X. Lim**, H.S. Greenside. The external magnetic field created by the superposition of identical parallel finite solenoids. American Journal of Physics **84**(8), 606–615, 2016

Peer-reviewed journal articles authored by mentees

- A.G. Kline, **M.X. Lim**, H.M. Jaeger. Precision measurement of tribocharging in acoustically levitated sub-millimeter grains. Review of Scientific Instruments, **91**(2), 023908, 2020
- I.A. Harris, **M.X. Lim**, H.M. Jaeger. Temperature dependence of nylon and PTFE triboelectrification. Phys. Rev. Materials, **3**(8), 085603, 2019

Review Articles

- **M.X. Lim**, B. vanSaders, H.M. Jaeger. Acoustic manipulation of multi-body structures and dynamics, Reports on Progress in Physics **87**(6), 2024.

Patents

- Portable spirometer and method for monitoring lung function G Poore, D Chander, Jinsu Kim, **M.X. Lim** - US Patent 10,098,570, 2018

In Submission

- **M.X. Lim**, Z. Liang, G. Alkuino, J. Z. Kim, I. Griniasty, T. Zhang, P.L. McEuen, I. Cohen. Multistable energy landscapes for autonomous microscopic machines.

- A. Xia, **M.X. Lim**, J.Z. Kim, B. vanSaders, H.M. Jaeger. Discovering the emergent nonlinear dynamics of acoustically levitated cube clusters.

REFERENCES

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Paul L. McEuen

plm23@cornell.edu